

Lukakline™
Montelukast sodium

QUALITATIVE AND QUANTITATIVE COMPOSITION

Lukakline Chewable Tablet 4 mg

Each uncoated tablet contains Montelukast sodium equivalent to Montelukast 4mg.

Lukakline Chewable Tablet 5 mg

Each uncoated tablet contains Montelukast sodium equivalent to Montelukast 5mg.

Lukakline Tablet 10 mg

Each film-coated tablet contains Montelukast sodium equivalent to Montelukast 10mg.

Excipients

Lukakline Chewable Tablet 4 mg and 5 mg

Mannitol, Hydroxypropyl cellulose (Klucel EXF), Croscarmellose sodium, Ferric oxide (Iron Oxide red), Isopropyl alcohol, Cherry flavour ART#594, Modified food starch, Aspartame, Cellulose microcrystalline (Avicel PH112), Magnesium stearate.

Lukakline Tablet 10 mg

Lactose monohydrate, Cellulose microcrystalline (Avicel PH 101), Croscarmellose sodium, Hydroxypropyl cellulose (Klucel LF), Cellulose microcrystalline (Avicel PH 112), Magnesium stearate, HPMC 2910/Hypromellose 3cps, Titanium dioxide, Macrogol 400, Iron oxide yellow, Iron oxide red, Iron oxide black.

PHARMACEUTICAL FORM

Lukakline Chewable Tablet 4 mg

Light pink to pink coloured, speckled oval shaped, biconvex tablets debossed with "MTS" on one side and "4" on the other side.

Lukakline Chewable Tablet 5 mg

Light pink to pink coloured, speckled round shaped, biconvex tablets debossed with "MTS" on one side and "5" on the other side.

Lukakline Tablet 10 mg

Brown coloured rounded square shaped, film-coated tablets debossed with "MTS" on one side and "10" on the other side.

CLINICAL INFORMATION

Indications

Lukakline is indicated in adults and pediatric patients 2 years of age and older for the:

- prophylaxis and chronic treatment of asthma, including the prevention of day- and night-time symptoms, the treatment of aspirin-sensitive asthmatic patients, and the prevention of exercise-induced bronchoconstriction.
- relief of day-time and night-time symptoms of allergic rhinitis (seasonal and perennial allergic rhinitis).

Dosage and Administration

The therapeutic effect of montelukast on parameters of asthma control occurs within one day. Patients should be advised to continue taking montelukast even if their asthma is under control, as well as during periods of worsening asthma.

This product should not be used concomitantly with other products containing the same active ingredient i.e. montelukast.

No dosage adjustment is necessary for pediatric patients, the elderly, for patients with renal insufficiency, or mild-to-moderate hepatic impairment, or for patients of either gender.

Montelukast sodium, 4 mg, chewable tablet

Montelukast sodium, 5 mg, chewable tablet

The tablets are to be chewed before swallowing. If taken in connection with food, montelukast chewable tablets should be taken 1 hour before or 2 hours after food.

Montelukast sodium, 10 mg, film-coated tablet

Montelukast film coated tablets can be taken with or without food.

Therapy with Montelukast in Relation to Other Treatments for Asthma

Montelukast can be added to a patient's existing treatment regimen.

Reduction in Concomitant Therapy:

Bronchodilator Treatments: Montelukast can be added to the treatment regimen of patients who are not adequately controlled on bronchodilator alone. When a clinical response is evident (usually after the first dose), the patient's bronchodilator therapy can be reduced as tolerated.

Inhaled Corticosteroids: Treatment with montelukast provides additional clinical benefit to patients treated with inhaled corticosteroids. A reduction in the corticosteroid dose can be made as tolerated. The dose should be reduced gradually with medical supervision. In some patients, the dose of inhaled corticosteroids can be tapered off completely. Montelukast should not be abruptly substituted for inhaled corticosteroids (see section Warnings and Precautions).

Route of Administration

For oral use.

Montelukast should be taken once daily. For asthma, the dose should be taken in the evening. For allergic rhinitis, the time of administration may be individualised to suit patient needs. Patients with both asthma and allergic rhinitis should take only one tablet daily in the evening.

Adults

Adults 15 Years of Age and Older with Asthma and/or Allergic Rhinitis

The dosage for adults and adolescents 15 years of age and older is one 10 mg tablet daily.

Children

This medicinal product is to be given to a child under adult supervision.

Children 6-14 years

Children 6 to 14 Years of Age with Asthma and/or Allergic rhinitis

The dosage for pediatric patients 6 to 14 years of age is one 5 mg chewable tablet daily.

Children 2 to 5 Years of Age with Asthma and/or Allergic rhinitis

The dosage for pediatric patients 2 to 5 years of age is one 4 mg chewable tablet daily. No dosage adjustment within this age group is necessary. 4 mg chewable tablet formulation is not recommended below 2 years of age.

Contraindications

- hypersensitivity to the active substance or to any of the excipients.

Warnings and Precautions

The efficacy of oral montelukast for the treatment of acute asthma attacks has not been established. Therefore, oral montelukast should not be used to treat acute asthma attacks. Patients should be advised to have appropriate rescue medication available.

Whilst the dose of concomitant inhaled corticosteroid may be reduced gradually under medical supervision, montelukast should not be abruptly substituted for inhaled or oral corticosteroids.

Neuropsychiatric events have been reported in patients taking montelukast (see Adverse Reactions). Since other factors may have contributed to these events, it is not known if they are related to montelukast. Physicians should discuss these adverse experiences with their patients and/or caregivers. Patients and/or caregivers should be instructed to notify their physician if these changes occur.

Montelukast should not be used as monotherapy for the treatment and management of exercise-induced bronchospasm. Patients who have exacerbations of asthma after exercise should continue to use their usual regimen of inhaled β -agonists as prophylaxis and have available for rescue a short-acting inhaled β -agonist.

The reduction in systemic corticosteroid dose in patients receiving anti-asthma agents including leukotriene receptor antagonists has been followed in rare cases by the occurrence of one or more of the following: eosinophilia, vasculitic rash, worsening pulmonary systems, cardiac complications and/or neuropathy sometimes diagnosed as Churg-Strauss syndrome, a systemic eosinophilic vasculitis. Although a causal relationship with leukotriene receptor antagonism has not been established, caution and appropriate clinical monitoring are recommended when systemic corticosteroid reduction is considered in patients receiving montelukast.

Treatment with montelukast does not alter the need for patients with aspirin-sensitive asthma to avoid taking aspirin and other non-steroidal anti-inflammatory drugs.

Lukakline 10 mg film-coated tablets contain lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Lukakline 4 mg and 5 mg chewable tablets contain aspartame, a source of phenylalanine. This may be harmful for people with phenylketonuria.

Interactions

Montelukast may be administered with other therapies routinely used in the prophylaxis and chronic treatment of asthma and in the treatment of allergic rhinitis. In drug-interactions studies, the recommended clinical dose of montelukast did not have clinically important effects on the pharmacokinetics of the following medicinal products: theophylline, prednisone, prednisolone, oral contraceptives (ethinyl oestradiol/norethindrone 35/1), terfenadine, digoxin and warfarin.

The area under the plasma concentration-time curve (AUC) for montelukast was decreased approximately 40% in subjects with co-administration of phenobarbital. No dosage adjustment for montelukast is recommended. Since montelukast is metabolised by CYP 3A4, 2C8, and 2C9 caution should be exercised, particularly in children, when montelukast is co-administered with inducers of CYP 3A4, 2C8, and 2C9, such as phenytoin, phenobarbital and rifampicin.

In vitro studies have shown that montelukast is a potent inhibitor of CYP 2C8. However, data from a clinical drug-drug interaction study involving montelukast and rosiglitazone (a probe substrate representative of drugs primarily metabolised by CYP 2C8) demonstrated that montelukast does not inhibit CYP 2C8 *in vivo*. Therefore, montelukast is not anticipated to alter the metabolism of drugs metabolised by this enzyme (e.g. paclitaxel, rosiglitazone, and repaglinide).

In vitro studies have shown that montelukast is a substrate of CYP 2C8, and to a less significant extent, of 2C9, and 3A4. In a clinical drug-drug interaction study involving montelukast and gemfibrozil (an inhibitor of both CYP 2C8 and 2C9) gemfibrozil increased the systemic exposure of montelukast by 4.4-fold. No routine dosage adjustment of montelukast is required upon co-administration with gemfibrozil or other potent inhibitors of CYP 2C8, but the physician should be aware of the potential for an increase in adverse reactions.

Based on *in vitro* data, clinically important drug interactions with less potent inhibitors of CYP 2C8 (e.g., trimethoprim) are not anticipated.

Co-administration of montelukast with itraconazole, a strong inhibitor of CYP 3A4, resulted in no significant increase in the systemic exposure of montelukast.

Pregnancy and Lactation

Pregnancy

Montelukast has not been studied in pregnant women. Montelukast should be used during pregnancy only if it is considered to be clearly essential.

During worldwide marketing experience, congenital limb defects have been rarely reported in the offspring of women being treated with montelukast during pregnancy. Most of these women were also taking other asthma medications during their pregnancy. A causal relationship between these events and montelukast has not been established.

Lactation

It is not known if montelukast is excreted in human milk. Because many drugs are excreted in human milk, caution should be exercised when montelukast is given to a nursing mother.

Pediatric Use

Montelukast has been studied in pediatric patients 6 months to 14 years of age. Safety and effectiveness in pediatric patients younger than 6 months of age have not been studied. Studies have shown that montelukast does not affect the growth rate of pediatric patients.

Adverse Reactions

Montelukast has been generally well tolerated. Adverse reactions, which usually were mild, generally did not require discontinuation of therapy. The overall incidence of adverse reactions reported with montelukast is comparable to placebo.

Clinical Trial Data

Adults 15 Years of Age and Older with Asthma

Montelukast has been evaluated in approximately 2600 adult patients 15 years of age and older in clinical studies. In two similarly designed, 12-week placebo-controlled clinical studies, the only adverse experiences reported as drug related in $\geq 1\%$ of patients treated with montelukast and at a greater incidence than in patients treated with placebo were abdominal pain and headache. The incidences of these events were not significantly different in the two treatment groups. Cumulatively, 544 patients were treated with montelukast for at least 6 months, 253 for one year and 21 for 2 years in clinical studies. With prolonged treatment, the adverse experience profile did not change.

Pediatric Patients 6 to 14 Years of Age with Asthma

Montelukast has been evaluated in approximately 475 pediatric patients 6 to 14 years of age. The safety profile in pediatric patients is generally similar to the adult safety profile and to placebo. In an 8-week, placebo-controlled clinical study, the only adverse experience reported as drug related in $> 1\%$ of patients treated with montelukast and at a greater incidence than in patients treated with placebo was headache. The incidence of headache was not significantly different in the two treatment groups.

In studies evaluating growth rate, the safety profile in these pediatric patients was consistent with the safety profile previously described for montelukast.

Cumulatively, 263 pediatric patients 6 to 14 years of age were treated with montelukast for at least 3 months and 164 for 6 months or longer. With prolonged treatment, the adverse experience profile did not change.

Pediatric Patients 2 to 5 Years of Age with Asthma

Montelukast has been evaluated in 573 pediatric patients 2 to 5 years of age. In a 12-week, placebo-controlled clinical study, the only adverse experience reported as drug related in $> 1\%$ of patients treated with montelukast and at a greater incidence than in patients treated with placebo was thirst. The incidence of thirst was not significantly different in the two treatment groups.

Cumulatively, 426 pediatric patients 2 to 5 years of age were treated with montelukast for at least 3 months, 230 for 6 months or longer, and 63 patients for 12 months or longer. With prolonged treatment, the adverse experience profile did not change.

Adults 15 Years of Age and Older with Seasonal Allergic Rhinitis

Montelukast has been evaluated in 2199 adult patients 15 years of age and older for the treatment of seasonal allergic rhinitis in clinical studies. Montelukast administered once daily in the morning or in the evening was generally well tolerated with a safety profile similar to that of placebo. In placebo-controlled, clinical studies, no adverse experiences reported as drug related in $\geq 1\%$ of patients treated with montelukast and at a greater incidence than in patients treated with placebo were observed. In a 4-week, placebo-controlled clinical study, the safety profile was consistent with that observed in 2-week studies. The incidence of somnolence was similar to that of placebo in all studies.

Pediatric Patients 2 to 14 Years of Age with Seasonal Allergic Rhinitis

Montelukast has been evaluated in 280 pediatric patients 2 to 14 years of age for the treatment of seasonal allergic rhinitis in a 2-week, placebo-controlled, clinical study. Montelukast administered once daily in the evening was generally well tolerated with a safety profile similar to that of placebo. In this study, no adverse experiences reported as drug related in $\geq 1\%$ of patients treated with montelukast and at a greater incidence than in patients treated with placebo were observed.

Adults 15 Years of Age and Older with Asthma and Seasonal Allergic Rhinitis

Montelukast 10 mg film-coated tablets have been evaluated in approximately 400 asthmatic patients 15 years of age and older with seasonal allergic rhinitis. The safety profile in asthmatic patients with seasonal allergic rhinitis was consistent with that observed in patients with asthma.

Adults 15 Years of Age and Older with Perennial Allergic Rhinitis

Montelukast has been evaluated in 3235 adult and adolescent patients 15 years of age and older with perennial allergic rhinitis in two, 6-week, placebo-controlled, clinical studies. Montelukast administered once daily was generally well tolerated, with a safety profile consistent with that observed in patients with seasonal allergic rhinitis and similar to that of placebo. In these two studies, no adverse experiences reported as drug related in $\geq 1\%$ of patients treated with montelukast and at a greater incidence than in patients treated with placebo were observed. The incidence of somnolence was similar to that of placebo.

Pediatric Patients 2 to 14 Years of Age with Perennial Allergic Rhinitis

The safety in patients 2 to 14 years of age with perennial allergic rhinitis is supported by the established safety in patients 2 to 14 years of age with seasonal allergic rhinitis.

Pooled Analysis of Clinical Trials Experience

A pooled analysis of 41 placebo-controlled clinical studies (35 studies in patients 15 years of age and older, 6 studies in pediatric patients 6 to 14 years of age) was performed using a validated assessment method of suicidality. Among the 9929 patients who received montelukast and 7780 patients who received placebo in these studies, there was one patient with suicidal ideation in the group taking montelukast. There were no completed suicides, suicide attempts or preparatory acts toward suicidal behaviour in either treatment group.

A separate pooled analysis of 46 placebo-controlled clinical studies (35 studies in patients 15 years of age and older, 11 studies in pediatric patients 3 months to 14 years of age) assessing behaviour-related adverse experiences (BRAEs) was performed. Among the 11,673 patients who received montelukast and 8827 patients who received placebo in these studies, the frequency of patients with at least one BRAE was 2.73% in patients who received montelukast and 2.27% in patients who received placebo; the odds ratio was 1.12 (95% CI [0.93, 1.36]). The clinical trials included in these pooled analysis were not designed specifically to examine suicidality or BRAEs.

Post-Marketing Data

Adverse reactions are ranked under headings of frequency using the following convention:

Very common $\geq 1/10$

Common $\geq 1/100$ to $<1/10$

Uncommon $\geq 1/1000$ to $<1/100$

Rare $\geq 1/10000$ to $<1/1000$

Very rare $<1/10000$

Not known (cannot be estimated from the available data).

The information given below is based on data from clinical studies and on post-marketing experience.

Infections and infestations

Very common: upper respiratory infection¹

Blood and lymphatic system disorders

Rare: increased bleeding tendency

Immune system disorders

Uncommon: hypersensitivity reactions including anaphylaxis, very rarely hepatic eosinophilic infiltration

Psychiatric disorders

Uncommon: agitation including aggressive behaviour or hostility, depression, dream abnormalities including nightmares, insomnia, somnambulism, anxiety, psychomotor hyperactivity (including irritability, restlessness, tremor)¹

Rare: disturbance in attention, memory impairment

Very rare: hallucinations, disorientation, suicidal thinking and behavior (suicidality)

Nervous system disorders

Common: headache¹

Uncommon: dizziness, drowsiness, paraesthesia/hypoesthesia, seizure

Cardiac disorders

Rare: palpitations

Respiratory, thoracic and mediastinal disorders

Uncommon: epistaxis

Very rare: Churg-Strauss Syndrome (see Section Warnings and Precautions), pulmonary eosinophilia

Gastrointestinal disorders

Common: abdominal pain¹, diarrhoea¹, nausea¹, vomiting¹

Uncommon: dry mouth, dyspepsia

Hepatobiliary disorders

Common: elevated levels of serum transaminases (ALT and AST)

Very rare: hepatitis (including cholestatic, hepatocellular and mixed pattern liver injury), hepatic eosinophilic infiltration

Skin and subcutaneous tissue disorders

Common: rash¹

Uncommon: urticaria, pruritus

Rare: angioedema

Very rare: erythema multiforme, erythema nodosum

Musculoskeletal and connective tissue and bone disorders

Uncommon: arthralgia, myalgia including muscle cramps

General disorders and administration site conditions:

Common: thirst¹, pyrexia¹

Uncommon: asthenia/ fatigue, malaise, oedema

Injury, poisoning and procedural complications

Uncommon: bruising

¹ only Montelukast sodium, 5 mg, chewable tablets and Montelukast sodium, 10 mg, film-coated tablets

² only Montelukast sodium, 4 mg, chewable tablets and Montelukast sodium, 10 mg, film-coated tablets

³ only Montelukast sodium, 4 mg, chewable tablets

⁴ only Montelukast sodium, 4 mg, oral granules

¹ this adverse experience, reported as Very Common in the patients who received montelukast, was also reported as Very Common in the patients who received placebo in clinical trials.

² This adverse experience, reported as Common in the patients who received montelukast, was also reported as Common in the patients who received placebo in clinical trials.

³ Frequency Category: Rare

The following side effects have also been reported in post-marketing use:

Psychiatric disorders: disturbance in attention, memory impairment, anxiety

Respiratory, thoracic and mediastinal disorders: pulmonary eosinophilia

Skin and subcutaneous tissue disorders: Stevens-Johnson Syndrome, toxic epidermal necrolysis

Overdosage

No specific information is available on the treatment of overdosage with montelukast. In chronic asthma studies, montelukast has been administered at doses up to 200 mg/day to adult patients for 22 weeks and in short-term studies, up to 900 mg/day to patients for approximately one week without clinically important adverse experiences.

There have been reports of acute overdosage in post-marketing experience and clinical studies with montelukast. These include reports in adults and children with a dose as high as 1000 mg. The clinical and laboratory findings observed were consistent with the safety profile in adults and pediatric patients. There were no adverse experiences in the majority of overdosage reports.

Symptoms and signs

The most frequently occurring adverse experiences were consistent with the safety profile of montelukast and included abdominal pain, somnolence, thirst, headache, vomiting, and psychomotor hyperactivity.

Treatment

It is not known whether montelukast is dialysable by peritoneal or haemodialysis.

Clinical Pharmacology

Pharmacodynamics

Pharmacotherapeutic group

Leukotriene receptor antagonist.

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	Market: APAC	List of Colours:								
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ATC Code

R03DC03

Mechanism of Action and Pharmacodynamic effects

The **cysteinyl leukotrienes** (LTC₄, LTD₄, LTE₄) are potent inflammatory eicosanoids released from various cells including mast cells and eosinophils. These important pro-asthmatic mediators bind to cysteinyl leukotriene (CysLT) receptors. The CysLT type-1 (CysLT₁) receptor is found in the human airway (including airway smooth muscle cells and airway macrophages) and on other pro-inflammatory cells (including eosinophils and certain myeloid stem cells). CysLTs have been correlated with the pathophysiology of asthma and allergic rhinitis. In asthma, leukotriene-mediated effects include bronchoconstriction, mucous secretion, vascular permeability and eosinophil recruitment. In allergic rhinitis, CysLTs are released from the nasal mucosa after allergen exposure during both early- and late-phase reactions and are associated with symptoms of allergic rhinitis. Intranasal challenge with CysLTs has been shown to increase nasal airway resistance and symptoms of nasal obstruction.

Montelukast is a potent, orally active compound with anti-inflammatory properties which significantly improves parameters of asthmatic inflammation. Based on biochemical and pharmacological bioassays, it binds with high affinity and selectivity to the CysLT₁ receptor (in preference to other pharmacologically important airway receptors such as the prostanoid, cholinergic or β -adrenergic receptor). Montelukast potently inhibits physiologic actions of LTC₄, LTD₄ and LTE₄ at the CysLT₁ receptor without any agonist activity.

In asthmatic patients, montelukast causes potent inhibition of airway cysteinyl leukotriene receptors as demonstrated by the ability to inhibit bronchoconstriction due to inhaled LTD₄.

Doses as low as 5 mg cause substantial blockage of LTD₄-induced bronchoconstriction. Montelukast causes bronchodilation within 2 hours of oral administration; these effects were additive to the bronchodilation caused by a β -agonist.

Clinical Studies – Asthma

In clinical studies, montelukast is effective in adult and pediatric patients for the prophylaxis and chronic treatment of asthma, including the prevention of day- and nighttime symptoms, the treatment of aspirin-sensitive asthmatic patients, and the prevention of exercise-induced bronchoconstriction. Montelukast is effective alone or in combination with other agents used in the maintenance treatment of chronic asthma.

Montelukast and inhaled corticosteroid may be used concomitantly with additive effects to control asthma or to reduce the inhaled corticosteroid dose while maintaining clinical stability.

Adults 15 years of age and older

In two similarly-designed 12-week double-blind placebo-controlled studies in adult asthmatic patients 15 years of age and older, montelukast, 10 mg once daily in the evening, demonstrated significant improvements in parameters of asthma control measuring asthma symptoms, asthma-related outcomes, respiratory function and "as needed" β -agonist use.

Montelukast significantly improved patient-reported daytime symptoms and nocturnal awakenings, compared with placebo. Asthma-specific outcomes, including asthma attacks, corticosteroid rescue, discontinuations due to worsening asthma, asthma exacerbations and asthma-free days were also significantly better than placebo.

Physicians' and patients' global asthma evaluations and asthma-specific quality-of-life evaluations (in all domains, including normal daily activity and asthma symptoms) were significantly better than placebo. Montelukast caused significant improvements in morning forced expiratory volume in 1 second (FEV₁), AM and PM peak expiratory flow rate (PEFR) and significantly decreased the use of "as-needed" β -agonist, compared with placebo.

The treatment effect was achieved after the first dose and was maintained throughout the 24-hour dosing interval. Treatment effect also remained constant during continuous once daily administration in extension studies for up to one year. Withdrawal of montelukast after 12 weeks of use did not cause rebound worsening of asthma.

Compared with inhaled beclomethasone (200 μ g twice daily with a spacer device), montelukast demonstrated a more rapid initial response although over the full duration of the 12-week study, beclomethasone provided a greater average treatment effect. However, a high percentage of patients treated with montelukast achieved similar clinical responses compared with inhaled beclomethasone.

Pediatric patients 6 to 14 years of age

In pediatric patients 6 to 14 years of age, one 5-mg chewable tablet daily in the evening, significantly decreased asthma exacerbations, and improved parents' global evaluations and the pediatric asthma-specific quality-of-life evaluations, compared with placebo.

Montelukast also significantly improved morning FEV₁ and decreased total daily "as needed" β -agonist use. Treatment effect was achieved after the first dose and remained constant during once-daily administration for up to 6 months.

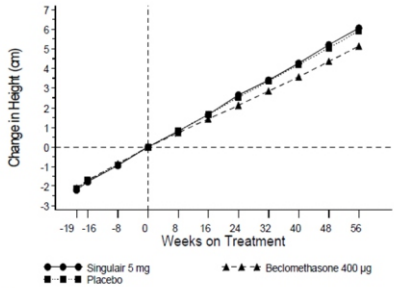
Growth Rate in Pediatric Patients

Two controlled clinical studies have demonstrated that montelukast did not affect the growth rate of prepubertal pediatric patients with asthma.

Growth rate was evaluated in a double-blind study of 71 pediatric patients aged 6 to 11 years (67 completed the study and were evaluated for growth): 35 patients were treated in a crossover fashion with montelukast and placebo; 32 patients were treated in a crossover fashion with inhaled budesonide and placebo. Growth rate as measured by lower leg length growth was similar in patients treated with MONTELUKAST 5 mg once daily for 3 weeks compared with placebo, whereas growth rate was significantly lower ($p=0.002$) in patients treated with inhaled budesonide dosed at 200 mcg (one puff of 200 mcg) twice daily for 3 weeks, compared with placebo.

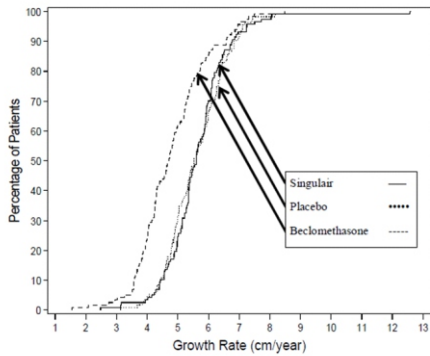
In a 56-week, multi-center, double-blind, randomized, active- and placebo-controlled parallel group study the effect of montelukast on growth rate was evaluated in 360 patients aged 6 to 8 years; 120 patients treated with montelukast. The observed mean linear growth rates in the placebo run-in period were similar in the three treatment arms. The mean growth rate in patients treated with montelukast 5 mg once daily was similar to that of placebo during the treatment period, while the mean growth rate in patients treated with inhaled beclomethasone dipropionate dosed at 200 mcg (two puffs of 100 mcg ex-valve) twice daily with a spacer device was significantly below the mean growth rate of patients receiving montelukast ($p<0.001$) or placebo ($p<0.001$) (see FIGURE 1). Across the distribution of growth rates during treatment, growth rates for the beclomethasone group were reduced compared to those of montelukast. The montelukast and placebo groups had similar growth rates (see FIGURE 2). Both montelukast and inhaled beclomethasone demonstrated significant benefit versus placebo in the exploratory efficacy endpoints of rescue medication use in these patients with mild asthma.

FIGURE 1
Change in Height (cm) from Randomization Visit by Scheduled Week
(Treatment Group Mean $\pm$ Standard Error[†] of the Mean)



[†] The standard errors of the treatment group means in change in height are too small to be visible on the plot.

FIGURE 2
Cumulative Percentage of Patients
by Growth Rate (cm/year) Over 56 Weeks of Treatment



Pediatric patients 6 months to 5 years of age

In a 12-week, placebo-controlled study in pediatric patients 2 to 5 years of age, montelukast 4 mg once daily consistently improved parameters of asthma control irrespective of concomitant controller therapy use compared with placebo. Sixty percent of patients were not on any other controller therapy. Montelukast significantly improved daytime symptoms (including coughing, wheezing, trouble breathing and activity limitation) and nighttime symptoms compared with placebo. Montelukast also significantly decreased "as-needed" β -agonist use and corticosteroid rescue compared with placebo. Patients receiving montelukast had significantly more days without asthma than those receiving placebo. A treatment effect was achieved after the first dose. In addition, total blood eosinophil counts were significantly decreased.

Efficacy of montelukast is supported in pediatric patients 6 months to 2 years of age by extrapolation from the demonstrated efficacy in patients 2 years of age and older with asthma, and is based on similar pharmacokinetic data, as well as the assumption that the disease course, pathophysiology and the drug's effect are substantially similar among these populations.

Effects in patients on concomitant inhaled corticosteroids

Separate studies in adults demonstrated the ability of montelukast to add to the clinical effect of inhaled corticosteroid and allow steroid tapering when used concomitantly. In a placebo-controlled study, patients taking initial inhaled corticosteroid doses of approximately 1600 μ g per day reduced their steroid use by approximately 37% during a placebo run-in period. Montelukast allowed a further 47% reduction of the inhaled corticosteroid dose, compared with 30% for placebo. In another study, montelukast provided additional clinical benefit to a similar population of patients maintained but not adequately controlled on inhaled corticosteroid (beclomethasone 400 μ g per day).

Complete abrupt removal of beclomethasone in patients receiving both montelukast and beclomethasone caused clinical deterioration in some patients, suggesting that tapering as tolerated rather than abrupt removal is preferred. In aspirin-sensitive patients, nearly all of whom were receiving concomitant inhaled and/or oral corticosteroids, montelukast resulted in significant improvement in the parameters of asthma control.

Effects on exercise-induced bronchoconstriction

Montelukast, 10 mg once daily, prevented exercise-induced bronchoconstriction (EIB) in adults

15 years of age and older. In a 12-week study, montelukast significantly inhibited the extent and duration of fall in FEV₁ over 60 minutes after exercise, the maximal percent fall in FEV₁ after exercise, and the time to recovery to within 5% of the pre-exercise FEV₁.

Protection was consistent through the treatment period indicating that tolerance did not occur. In a separate cross-over study, protection was observed after two once-daily doses. In pediatric patients 6 to 14 years of age, using the 5-mg chewable tablet, a similarly designed cross-over study demonstrated similar protection and the protection was maintained throughout the dosing interval (24 hours).

Effects on asthmatic inflammation

In clinical studies montelukast inhibited both early- and late-phase bronchoconstriction due to antigen challenge. Because inflammatory cell (eosinophil) infiltration is an important feature of asthma, the effects of montelukast on eosinophils in the peripheral blood and airway were examined. In Phase IIb/III clinical studies, montelukast significantly decreased peripheral blood eosinophils approximately 15% from baseline, compared with placebo. In pediatric patients 6 to 14 years of age, montelukast decreased peripheral blood eosinophils 13% over the 8-week treatment period, compared with placebo.

Montelukast also significantly decreased airway eosinophils in sputum, compared with placebo. In this study, peripheral blood eosinophils decreased and clinical asthma endpoints improved with treatment with Montelukast.

Effects in Patients with Asthma and Seasonal Allergic Rhinitis

In a clinical study in adult asthmatic patients 15 years of age and older with concomitant seasonal allergic rhinitis, montelukast 10-mg tablets administered once daily demonstrated a statistically significant improvement in the primary variable, Daily Rhinitis Symptoms score (average of the Daytime Nasal Symptoms score [mean of nasal congestion, rhinorrhea, sneezing, nasal itching] and the Nighttime Symptoms score [mean of nasal congestion upon awakening, difficulty going to sleep, and nighttime awakenings scores]), compared with placebo. Global evaluations of allergic rhinitis by patients and physicians, and global evaluations of asthma by patients and physicians, were also significantly improved, compared with placebo.

Clinical Studies - Allergic Rhinitis

The efficacy of montelukast for the treatment of seasonal allergic rhinitis was investigated in similarly designed randomized, 2-week, double-blind, placebo-controlled trials including 4924 patients (1751 patients were treated with montelukast). Patients were 15 years of age and older with a history of seasonal allergic rhinitis, a positive skin test to at least one relevant seasonal allergen, and active symptoms of seasonal allergic rhinitis at study initiation.

In a combined analysis of three pivotal studies, montelukast 10-mg tablets administered to 1189 patients once daily in the evening resulted in a statistically significant improvement in the primary endpoint, daytime nasal symptoms score, and its individual components (nasal congestion, rhinorrhea, nasal itching and sneezing); nighttime symptoms score, and its individual components (nasal congestion upon awakening, difficulty going to sleep, and nighttime awakenings); composite symptoms score (composed of the daytime nasal and nighttime symptoms scores); and global evaluations of allergic rhinitis by patients and by physicians, compared with placebo.

In a separate 4-week study in which montelukast was administered once daily in the morning, the efficacy over the initial 2 weeks was significantly different from placebo and consistent with the effect observed in studies using evening dosing. Additionally, the effect over the entire 4 weeks was consistent with the 2-week results.

In patients with seasonal allergic rhinitis aged 15 years and older who received montelukast, a median decrease of 13% in peripheral blood eosinophil counts was noted, compared with placebo, over the double-blind treatment periods.

The efficacy of montelukast for the treatment of perennial allergic rhinitis was investigated in two, similarly designed randomized, 6-week, double-blind, placebo-controlled studies including 3235 patients (1632 patients were treated with montelukast). Patients were 15 to 82 years of age with a history of perennial allergic rhinitis, positive skin test results to relevant perennial allergens (including dust mites, animal dander, and mold spores), and active symptoms of perennial allergic rhinitis at study initiation.

In one study, montelukast 10-mg tablets administered to 1000 patients once daily resulted in a statistically significant improvement in the primary endpoint, Daytime Nasal Symptoms score, and its individual components (nasal congestion, rhinorrhea, and sneezing), compared with placebo. Montelukast also demonstrated patient-perceived improvement of allergic rhinitis as assessed by the secondary endpoints of Global Evaluation of Allergic Rhinitis by Patient, and Rhinoconjunctivitis Quality-of-Life overall score (average of scores for the 7 domains of activity, sleep, non-nose/non-eye symptoms, practical problems, nasal symptoms, eye symptoms, and emotions), compared with placebo.

The efficacy of montelukast for the treatment of seasonal allergic rhinitis in pediatric patients 2 to 14 years of age, and for the treatment of perennial allergic rhinitis in pediatric patients 1 year to 14 years of age is supported by extrapolation from the demonstrated efficacy in patients 15 years of age and older with allergic rhinitis as well as the assumption that the disease course, pathophysiology and the drug's effect are substantially similar among these populations.

Pharmacokinetics

Absorption

Montelukast is rapidly and nearly completely absorbed following oral administration. For the 10 mg film-coated tablet, the mean peak plasma concentration (C_{max}) is achieved 3 hours (T_{max}) after administration in adults in the fasted state. The mean oral bioavailability is 64%. The oral bioavailability and C_{max} are not influenced by a standard meal. Safety and efficacy were demonstrated in clinical trials where the 10 mg film-coated tablet was administered without regard to the timing of food ingestion.

For the 5 mg chewable tablet, C_{max} is achieved in 2 hours after administration in adults in the fasted state. The mean oral bioavailability is 73%. Food does not have a clinically important influence with chronic administration.

For the 4 mg chewable tablet, C_{max} is achieved in 2 hours after administration in pediatric patients 2 to 5 years of age in the fasted state.

Distribution

Montelukast is more than 99% bound to plasma proteins. The steady-state volume of distribution of montelukast averages 8-11 liters. Studies in rats with radiolabelled montelukast indicate minimal distribution across the blood-brain barrier. In addition, concentrations of radio-labelled material at 24 hours post-dose were minimal in all other tissues.

Metabolism

Montelukast is extensively metabolised. In studies with therapeutic doses, plasma concentrations of metabolites of montelukast are undetectable at steady state in adults and children. The contribution of metabolites to the therapeutic effect of montelukast is minimal. In studies with therapeutic doses, plasma concentrations of metabolites of montelukast are undetectable at steady state in adults and children. Cytochrome P450 2C8 is the major enzyme in the metabolism of montelukast. Additionally CYP 3A4 and 2C9 may have a minor contribution, although itraconazole, an inhibitor of CYP 3A4, was shown not to change pharmacokinetic variables of montelukast in healthy subjects that received 10 mg montelukast daily. Based on further *in vitro* results in human liver microsomes, therapeutic plasma concentrations of montelukast do not inhibit cytochromes P450 3A4, 2C9, 1A2, 2A6, 2C19 or 2D6.

Elimination

The plasma clearance of montelukast averages 45 ml/min in healthy adults. Following an oral dose of radio-labelled montelukast, 86% of the radioactivity was recovered in 5-day faecal collections and <0.2% was recovered in urine. Coupled with estimates of montelukast oral bioavailability, this indicates that montelukast and its metabolites are excreted almost exclusively via the bile.

In several studies, the mean plasma half-life of montelukast ranged from 2.7 to 5.5 hours in healthy young adults. The pharmacokinetics of montelukast are nearly linear for oral doses up to 50 mg. No difference in pharmacokinetics was noted between dosing in the morning or in the evening. During once-daily dosing with 10 mg montelukast, there is little accumulation of the parent drug in plasma (~14%).

Special patient populations

Elderly, Patients and Patients with Renal or Hepatic Insufficiency

No dosage adjustment is necessary for the elderly, for patients with renal insufficiency or mild to moderate hepatic insufficiency. There are no clinical data in patients with severe hepatic insufficiency (Child-Pugh score >9).

Pediatric Patients

In children 6 to 11 months of age, the systemic exposure to montelukast and the variability of plasma montelukast concentrations were higher than those observed in adults. Based on population analyses, the mean AUC was 60% higher and mean C_{max} was 89% higher than those observed in adults. The systemic exposure in children 12 to 23 months of age was less variable, but was still higher than that observed in adults. The mean AUC was 33% higher and the mean C_{max} was 60% higher than those observed in adults. Safety and tolerability of montelukast in a single-dose pharmacokinetic study in 26 children 6 to 23 months of age were similar to that of patients two years and above.

PHARMACEUTICAL INFORMATION

Shelf-Life

The expiry date is indicated on the packaging.

Storage

Store below 30°C.

Presentation

Lukakline Chewable Tablet 4 mg and 5 mg

Pack of 30 tablets.

Lukakline Tablet 10 mg

Pack of 28 and 30 tablets.

Nature and Contents of Container

Alu blister (with desiccant interlayer) with Alu foil.

Use and Handling

There are no special requirements for use or handling of this product.

Version number: SIN05 (GEX04/NCDS06)

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Manufactured by: Dr. Reddy's Laboratories Limited, Telangana, India.